

MedReview Project Report

A Proof-Verified Review Platform for Hospitals, Doctors & Medical Services

Team: Group 05 - THREE ALPHA

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Repository: github.com/Raihans-code/Med-review

Stack: Node.js, Express 5, PostgreSQL, Prisma, Zod

Executive Summary

MedReview is a relational database-backed review and directory platform designed to establish trust in healthcare services by providing proof-verified ratings for hospitals, doctors, and medical services. The project addresses the prevalent trust deficit in online healthcare reviews by requiring verifiable evidence—such as medical bills, appointment numbers, or supporting documents—before feedback is approved and published. The system utilizes a normalized PostgreSQL database schema managed through Prisma ORM and exposes functionality via an Express.js/Node.js REST API. User authentication, input validation with Zod, and role-based access control separate regular patient privileges from administrative oversight. The relational schema comprises 12 core models linked through primary keys, foreign keys, unique constraints, and enums to maintain referential integrity and support detailed healthcare analytics.

1. Introduction

- **Trust Deficit in Healthcare:** Patients frequently struggle to find reliable healthcare providers due to unverified, anonymous, or manipulated reviews across traditional directories.
- **Proof-Verified Feedback:** Every submitted review requires evidence (e.g., appointment number, bill, or file upload) before undergoing admin moderation.
- **Multi-Dimensional Metrics:** Instead of a single flat star rating, the platform collects distinct feedback on staff behavior, punctuality, doctor knowledge, and bedside manner.
- **Unified Provider Directory:** Integrates hospitals, doctors, specialties, and specialized medical services within a single relational system.
- **Target Beneficiaries:** Patients seeking verified medical care, platform administrators moderating content, and healthcare organizations aiming for authentic feedback.

2. Problem Identification

- **No Proof of Visit:** Anonymous and unverified feedback with no proof that an actual consultation occurred.
- **Manipulated Ratings:** Fake positive reviews falsely elevate clinics while unverified negative reviews damage practitioner reputations.
- **One Flat Rating Scale:** Standard 5-star scales fail to distinguish between operational efficiency (e.g., wait time) and clinical quality (e.g., bedside manner).

- **Dynamic Affiliations:** Doctors practice across multiple institutions; static directories fail to capture complex doctor-hospital relationships.
- **Weak Moderation & Privacy Risks:** Absence of structured review pipelines risks publishing offensive content or leaking sensitive patient data.
- **Lack of Auditability:** Absence of administrative logging hinders compliance and system security tracking.

3. Objectives

3.1 General Objectives

- Build patient trust using proof-verified submissions and transparent multi-dimensional rating metrics.
- Maintain data integrity across dynamic doctor, hospital, specialty, and service entity relationships.
- Enforce security, HIPAA-style privacy considerations, and administrative audit logging.

3.2 Administrator Objectives

- Verify and moderate submitted reviews by approving or rejecting submissions with internal notes.
- Manage directory entities including hospitals, doctors, specialties, medical services, and doctor-hospital affiliations.
- Review user reports concerning spam, offensive language, or HIPAA violations.
- Monitor security events and system actions via structured system logs.

3.3 Patient / User Objectives

- Search and browse verified providers filtered by specialty or geographic city.
- Submit multi-dimensional ratings backed by mandatory proof (bills, appointment IDs, or documents).
- Save/bookmark preferred providers for future access.
- Vote on the helpfulness of published reviews and flag non-compliant or fraudulent submissions.

4. Features

AREA	KEY FEATURES
Authentication & Users	JWT-based authentication, password hashing with bcrypt, role management (USER, ADMIN).
Provider Directory	CRUD operations for Hospitals, Doctors, Specialties, and Medical Services.
Affiliation Management	Dynamic Doctor-Hospital junction mappings supporting primary institution designations.
Proof-Backed Reviews	Multi-category ratings (staff, punctuality, knowledge, bedside manner) with mandatory file/ID proof.
Moderation Engine	Pending review queue for admins with approve/reject actions and logging notes.

User Engagement	Helpfulness voting (HELPFUL, NOT_HELPFUL), community content reporting, and saved listings.
Security & Auditing	Zod payload validation, role middleware, and masked IP address logging in SystemLog.

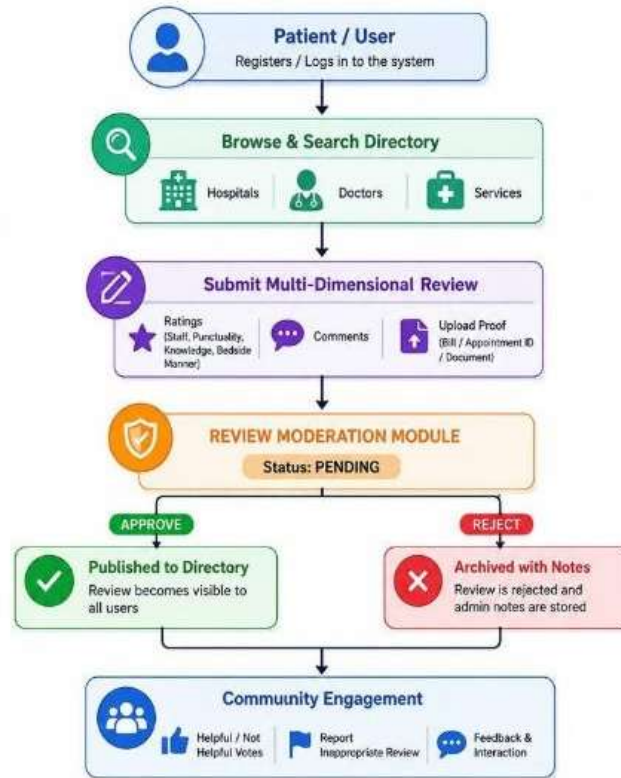
5. System Design Decisions & Edge Case Solutions

- **Pending Status by Default:** Every new review defaults to PENDING status to prevent unmoderated or fake feedback from going live.
- **Spam Prevention in Voting:** Unique compound constraint on [reviewId, userId] within the ReviewVote model prevents duplicate votes.
- **Flexible Target Entities:** Polymorphic-like optional foreign keys in Review allow feedback to directly target a Hospital, Doctor, or Medical Service.
- **Dynamic Doctor Affiliations:** Dedicated DoctorAffiliation join table with an isPrimary flag tracks doctors practicing at multiple hospitals.
- **Flexible Sub-Ratings:** Staff behavior and punctuality ratings are mandatory, while clinical metrics (knowledge, bedside manner) remain optional for general service reviews.
- **Privacy-Conscious Logging:** SystemLog ipAddress stores IP addresses in a masked, truncated format to uphold patient privacy standards.

6. System Workflow Diagram

Figure 1 illustrates the MedReview architectural and review moderation workflow from patient submission to public listing or administrative rejection:

MedReview – Review Workflow Diagram



7. System Architecture

LAYER	TECHNOLOGY / RESPONSIBILITY
Client/Consumer	Web/Mobile Clients or API Consumers sending HTTP requests.
Authentication & Validation	JWT Auth Middleware, bcrypt password hashing, Zod schema validation.
Application Layer	Node.js with Express.js 5 handling routes, business logic, and controllers.
Data Access Layer	Prisma ORM with @prisma/adapters-pg driver.
Database Layer	Relational PostgreSQL database environment.
Development Tools	Git, GitHub, TypeScript Types, dotenv, nodemon.

8. Entity Relationship Diagram (ERD) Structure

The core database architecture consists of 12 interconnected relational tables:



9. Database Schema & Enumerations

TABLE	PURPOSE	KEY RELATIONS
User	User profiles and credentials	1:M Review, SavedListing, ReviewVote, ReviewReport
Hospital	Hospital directory profiles	1:M DoctorAffiliation, MedicalService, Review
Doctor	Practitioner profiles	M:1 Specialty, 1:M DoctorAffiliation, Review
Specialty	Medical domain classifications	1:M Doctor
MedicalService	Specific treatments/services	M:1 Hospital, 1:M Review
DoctorAffiliation	Doctor-Hospital assignments	M:1 Doctor, Hospital
Review	Core review entry	M:1 User, Optional Target, 1:M ReviewProof
ReviewProof	Verification proof files/IDs	M:1 Review
ReviewReport	Flagged content queue	M:1 Review, User
ReviewVote	Review helpfulness tracking	M:1 Review, User
SavedListing	User bookmark list	M:1 User, Optional provider foreign key
SystemLog	System activity and audit trail	Optional M:1 User

9.1 Enumerations

ENUM NAME	ALLOWED VALUES
UserRole	USER, ADMIN
ProofType	BILL, APPOINTMENT_NUMBER, FILE_UPLOAD
ApprovalStatus	PENDING, APPROVED, REJECTED
ReportReason	SPAM, OFFENSIVE, HIPAA_VIOLATION, FALSE_INFO, OTHER
VoteType	HELPFUL, NOT_HELPFUL
LogSeverity	INFO, WARNING, ERROR, SECURITY

10. Requirement Tools

CATEGORY	TOOLS / TECHNOLOGIES
Runtime & Framework	Node.js, Express.js 5
Language & Types	JavaScript / TypeScript types
Database & ORM	PostgreSQL, Prisma ORM (@prisma/adapters-pg)
Security & Validation	JWT (JSON Web Tokens), bcrypt, Zod schema validator
Development Utilities	dotenv, nodemon, Git, GitHub

11. Implementation Steps

1. Conducted domain analysis on healthcare review trust deficits.
2. Designed normalized relational schema containing 12 entities.
3. Implemented Prisma Schema and connected PostgreSQL instance via Prisma Client.
4. Built authentication endpoints with bcrypt password hashing and JWT token management.
5. Created authorization middleware (authMiddleware, adminMiddleware) for access control.
6. Developed REST API routes/controllers for CRUD operations on directory entities.
7. Built the proof-backed review engine and admin moderation workflow.
8. Implemented user interaction features (SavedListings, ReviewVotes, and Content Reports).
9. Populated database with test seed data and verified role-based restrictions.

12. API Design & Sample Endpoints

The screenshot displays a REST client interface with the following details:

- Endpoint:** POST `{{baseUrl}}/reviews`
- Request Body (raw):**

```

1 {
2   "hospitalId": "{{hospitalId}}",
3   "doctorId": "{{doctorId}}",
4   "serviceId": "{{serviceId}}",
5   "staffRating": 4,
6   "punctualityRating": 5,
7   "knowledgeRating": 4,
8   "bedsideRating": 5,
9   "overallRating": 4,
10  "comment": "Not so bad."
11 }

```
- Response:** 201 Created, 517 ms, 801 B
- Response Body (JSON):**

```

{
  "overallRating": 4,
  "comment": "Not so bad.",
  "status": "PENDING",
  "adminNotes": null,
  "createdAt": "2026-08-04T00:17:12.186Z",
  "updatedAt": "2026-08-04T00:17:12.186Z",
  "doctorId": "e5792cdf-ft51-4840-89as-75b0833e242f",
  "serviceId": "82ec522e-51a7-4023-8174-11ad2c77ecdd"
}

```


FilesMed ReviewPOST Add R...PORT LoginGET My Rev...PUT Update U...GET Get Rev...POST Regis...POST Create...+Med Review - Local

Med Review API > Reviews > Add Review ProofSaveShare

POST((baseUrl)) /reviews/b766398b-d496-4091-9764-89a0249f3184/proofSend

DocsParamsAuthorizationHeaders 11BodyScriptsSettingsCookies

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSONSchemaBeautify

```
1 {
2   "type": "FILE_UPLOAD",
3   "proofValue": "https://example.com/proof.jpg"
4 }
```

BodyCookiesHeadersTest Results201 Created · 477 ms · 519 BSave Response

{ } JSONPreviewVisualize

proofId	64ec9512-4934-4a05-8233-a7ac36024ceb
reviewId	b766398b-d496-4091-9764-89a0249f3184
type	FILE_UPLOAD
proofValue	https://example.com/proof.jpg
uploadedAt	2026-08-04T00:18:24.570Z

GlobalVaultTools

Files Med Rev POST Add POST Login GET My Rx PUT Update GET Get R POST Reg POST Cnt POST Vote + Med Review - Local

Med Review API > Reviews > **Vote Review** Save Share

POST {{baseUrl}}/reviews/b766398b-d496-4091-9764-89a0249f3184/vote Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```

1 {
2   "voteType": "HELPFUL"
3 }

```

Body Cookies 1 Headers 11 Test Results 201 Created · 552 ms · 520 B Save Response

{ } JSON Preview Visualize

voteId	1d3de4ee-dcd8-42a2-8271-603ec0596528
reviewId	b766398b-d496-4091-9764-89a0249f3184
userId	dfb4232-8973-425b-b7b8-d928ae28e487
voteType	HELPFUL
createdAt	2026-08-04T00:20:15.385Z

Globals Vault Tools

Files Med Rev POST Add POST Login GET My F PUT Update GET Get T POST Reg POST Cnt POST Vot POST Rep + Med Review - Local

Med Review API > Reviews > **Report Review** Save Share

POST {{baseUrl}}/reviews/b766398b-d496-4091-9764-89a0249f3184/report Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```

1 {
2   "reason": "SPAM",
3   "details": "This review appears to be spam."
4 }

```

Body Cookies 1 Headers 11 Test Results 201 Created · 483 ms · 580 B Save Response

{ } JSON Preview Visualize

reportId	faaa9946-c658-4937-8ad6-8ccb8312397e
reviewId	b766398b-d496-4091-9764-89a0249f3184
userId	dfb4232-8973-425b-b7b8-d928ae28e487
reason	SPAM
details	This review appears to be spam.
isHandled	false
createdAt	2026-08-04T00:21:14.516Z

Globals Vault Tools

Med Review - Local

Med Review API > Reviews > My Reviews

GET {{baseUrl}}/reviews/MyReviews/ Send

Docs Params Authorization Headers Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers Test Results 200 OK 1.09 s 1.97 KB Save Response

JSON Preview Visualize

reviewid	userid	hospitalid	staffRating	punctualityRating	knowledgeRating	bedsideRating	overallRating	comment	status	adminNote
76c1ffa4-9b9b-41b0-b647-13a8dfaf3018	d928ae28e487	dd1535ea-7ddb-4a41-b7ac-ce7ebe86eeeb	4	5	4	5	4	Not so bad.	PENDING	null
b766398b-d496-4091-9764-89a0249f3184	d928ae28e487	dd1535ea-7ddb-4a41-b7ac-ce7ebe86eeeb	4	5	4	5	4	Great experience overall.	PENDING	null

Globals Vault Tools

Med Review - Local

Med Review API > Saved-Listings > Get My Saved Listings

GET {{baseUrl}}/saved-listings Send

Docs Params Authorization Headers Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers Test Results 200 OK 485 ms 521 B Save Response

JSON Preview Visualize

savedId	userid	entityId	createdAt	entityType
beb99478-69f6-405c-847d-6a11df133581	d928ae28e487	dd1535ea-7ddb-4a41-b7ac-ce7ebe86eeeb	2028-08-03T11:30:19.883Z	HOSPITAL

Globals Vault Tools

Med Re: POST Adc POST Login GET My F PUT Update GET Get POST Req POST Cre POST Vol POST Rep POST Sav + Med Review - Local

Med Review API Saved-Listings Save Listing Save Share

POST {{baseUrl}}/saved-listings Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "entityId": "74ea081a-c000-4183-abce-3849cfafae58",
3   "entityType": "HOSPITAL"
4 }
```

Body Cookies Headers Test Results 201 Created 477 ms 524 B Save Response

JSON Preview Visualize

savedId b9071972-1414-4afb-Bef1-3c00c67b4411

userId dfb4232-8973-425b-b7b0-d928ae28e487

entityId 74ea081a-c000-4183-abce-3849cfafae58

createdAt 2026-08-04T00:26:15.298Z

entityType HOSPITAL

Globals Vault Tools

ST Adc POST Log GET My F PUT Update GET Get POST Req POST Cre POST Vol POST Rep POST Sav PATCH Up + Med Review - Local

Med Review API Reviews Update Review Status Save Share

PATCH {{baseUrl}}/reviews/715a269d-218f-4a05-a218-513f440dd83/status Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "status": "APPROVED",
3   "adminNotes": "Verified and approved."
4 }
```

Body Cookies Headers Test Results 200 OK 134 ms 827 B Save Response

JSON Preview Visualize

overallRating 4

comment Not so bad.

status APPROVED

adminNotes Verified and approved.

createdAt 2026-08-04T01:22:24.105Z

updatedAt 2026-08-04T01:22:50.189Z

doctorId e5792cdf-f51-4840-89aa-75b0833e242f

serviceId 82ec522e-51a7-4023-8174-11ad2c77ecdd

Globals Vault Tools

GET My FPUT UpdateGET GetPOST ReqPOST CrePOST ValPOST ReqPOST SavPATCH UpDEL Dele+Med Review - Local

Med Review APIReviewsDelete ReviewSaveShare

DELETE{{baseUrl}}/reviews/09ab0299-c670-4d5f-a2ce-481bb9ef0cc2Send

DocsParamsAuthorizationHeadersBodyScriptsSettingsCookies

Query Params

Key	Value	Description
Key	Value	Description

BodyCookies 1Headers 11Test Results204 No Content101 ms350 BSave Response

RawPreviewVisualize

GET My FPUT UpdateGET GetPOST ReqPOST CrePOST ValPOST ReqPOST SavPATCH UpDEL DeleGET Get Lo+Med Review - Local

Med Review APISystem-LogsGet LogsSaveShare

GET{{baseUrl}}/system-logsSend

DocsParamsAuthorizationHeadersBodyScriptsSettingsCookies

Query Params

Key	Value	Description
severity		
userid		
action		
Key	Value	Description

BodyCookies 1Headers 11Test Results200 OK108 ms1.92 KBSave Response

JSONPreviewVisualize

	logId	userid	action	severity	details	ipAddress	createdAt	user	
0	a319ca2f-8bef-4992-98e6-8486aa896fd3	559818cf-f885-456d-a65b-220135be9d42	DELETE_REVIEW	WARNING	Review 09ab0299-c670-4d5f-a2ce-481bb9ef0cc2 deleted	1	2026-08-04T01:24:34.409Z	firstName	Raihan
								lastName	Islam
								email	iraihan705@gmail.com
1	ee48c44a-f8a0-4991-b790-8b72c2c19ff2	559818cf-f885-456d-a65b-220135be9d42	DELETE_REVIEW	WARNING	Review 715a269d-218f-4a05-a218-513f4f40dd83	1	2026-08-04T01:23:45.693Z	firstName	Raihan
								lastName	Islam
								email	iraihan705@gmail.com

DetailsVaultTools

My 1 • PUT Update • GET Get 1 • POST Req • POST Cre • POST Vot • POST Req • POST Sav • PATCH Up • DEL Dele • GET Get 1 • + • Med Review - Local

Med Review API • System-Logs • Get Log By Id

GET • {{baseUrl}} /system-logs/adec6f3e-e706-416f-969c-68c61042ebd6

Send

Docs Params Authorization Headers Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies 1 Headers 16 Test Results

200 OK • 370 ms • 701 B • Save Response

JSON Preview Visualize

logId adec6f3e-e706-416f-969c-68c61042ebd6

userId 6ed1db73-fe90-474c-bc1b-2c473bf9e35c

action REVIEW_APPROVED

severity INFO

details Admin approved a pending review

ipAddress 192.168.1.4

createdAt 2026-06-03T09:17:19.070Z

Globals Vault Tools

Update • GET Get 1 • POST Req • POST Cre • POST Vot • POST Req • POST Sav • PATCH Up • DEL Dele • GET Get 1 • DEL Dele • + • Med Review - Local

Med Review API • System-Logs • Delete Log

DELETE • {{baseUrl}} /system-logs/adec6f3e-e706-416f-969c-68c61042ebd6

Send

Docs Params Authorization Headers Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

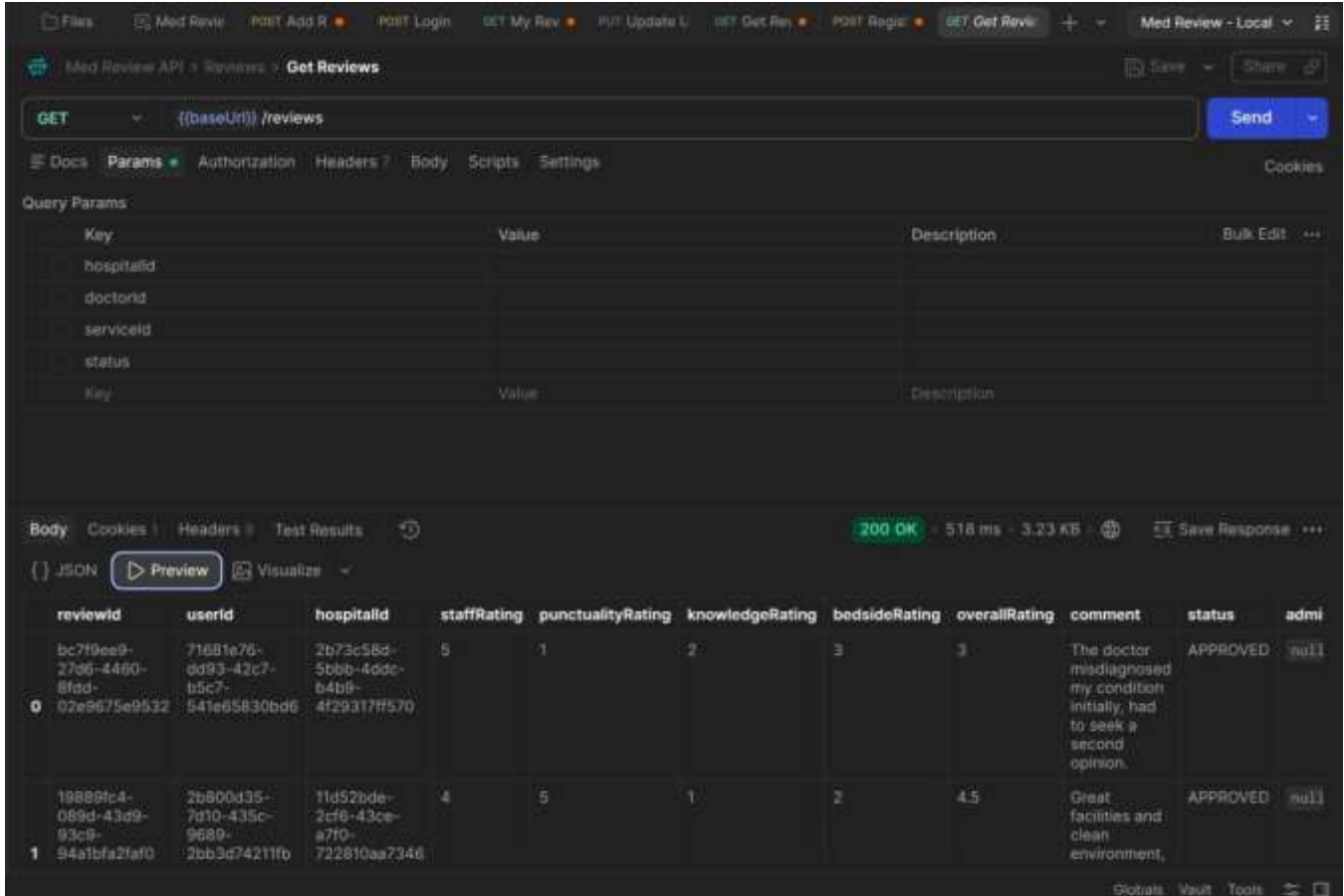
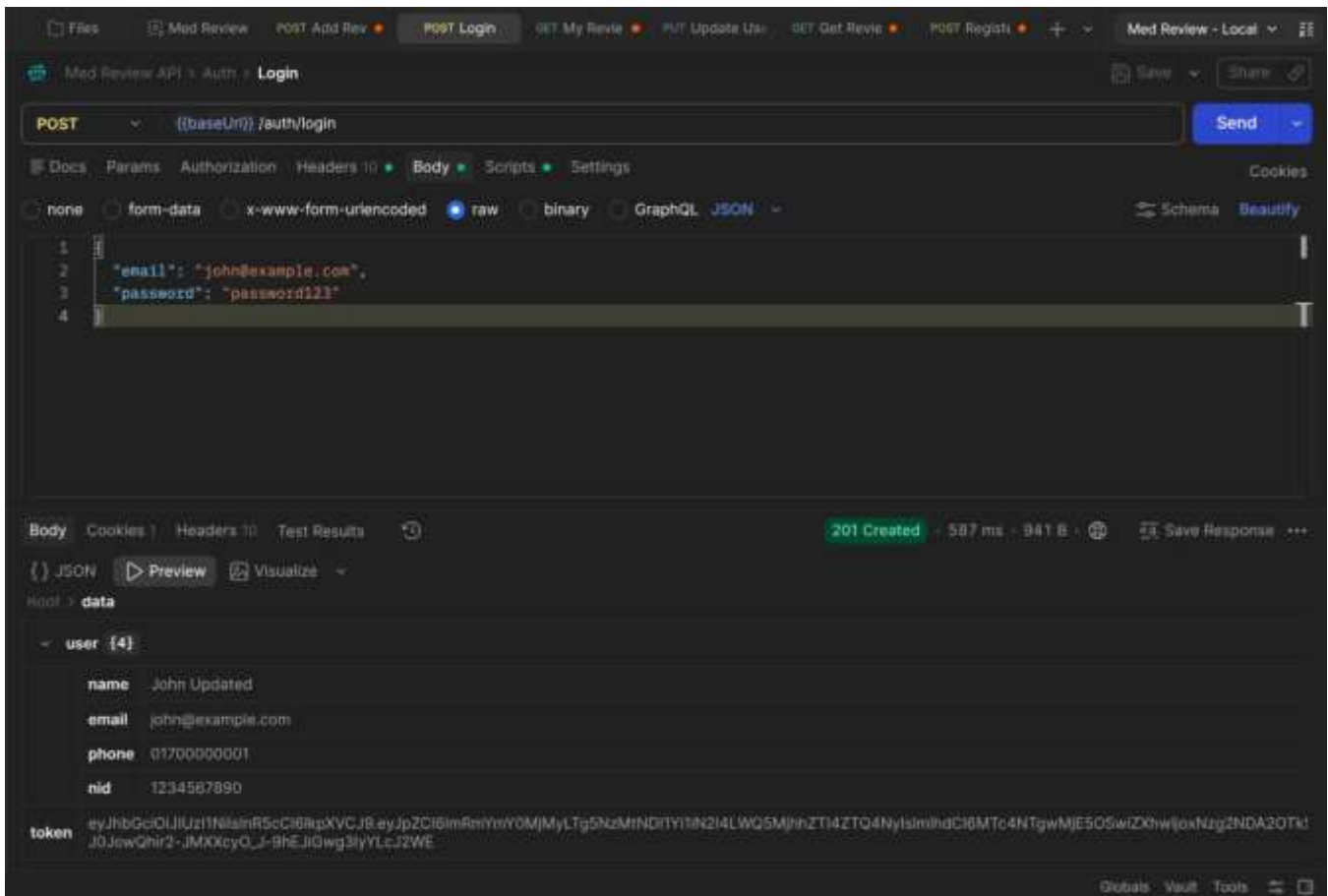
Body Cookies 1 Headers 11 Test Results

204 No Content • 94 ms • 350 B • Save Response

Raw Preview Visualize

Globals Vault Tools

Page 1 of Report • Proof-Verified Healthcare Review Platform



13. Data Analysis Potential

- **Moderation Metrics:** Tracking ratios of APPROVED vs. REJECTED reviews to quantify content credibility.
- **Report Breakdown:** Aggregating ReviewReport.reason entries to identify main sources of bad content (e.g., spam vs. HIPAA violations).
- **Affiliation Insights:** Mapping DoctorAffiliation patterns to evaluate practitioner distribution across healthcare facilities.
- **Helpfulness Trends:** Evaluating ReviewVote ratios to highlight reviews deemed most reliable by the patient community.
- **Dimensional Performance:** Comparing staff, punctuality, knowledge, and bedside manner scores to surface specific operational weaknesses in clinics.

14. Results

- **Functional Authentication:** Role-based JWT authentication and bcrypt password security fully implemented.
- **Directory Management:** Complete CRUD capabilities for Hospitals, Doctors, Specialties, and Medical Services.
- **Verification Architecture:** Review submission enforces mandatory proof attachment and defaults to PENDING.
- **Admin Control:** Functional moderation system allows admins to review proof, attach notes, and approve/reject feedback.
- **Community Interactions:** Helpfulness voting, reporting workflows, and user saved listings are operational.

15. Testing and Validation

TEST AREA	EXPECTED RESULT
Unique Voting Check	Duplicate votes by the same user on a single review are rejected.
Zod Input Validation	Invalid payload bodies or missing required sub-ratings return 400 Bad Request.
Role Authorization	Non-admin users attempting to access /api/admin/ receive 403 Forbidden.
Foreign Key Safety	Reviews targeting non-existent doctors or hospitals are rejected by database constraints.
Proof Enforceability	Review submission fails if no corresponding proof object is provided.

16. Challenges Encountered

- **Multi-Target Relational Structure:** Designing a flexible review schema capable of cleanly referencing doctors, hospitals, or services without breaking normalization.

- **Dynamic Doctor Affiliations:** Managing many-to-many linkages between doctors and hospitals while tracking primary workplaces.
- **Data Privacy in Logs:** Stripping sensitive information and masking IP addresses within audit logs.
- **Sub-Rating Granularity:** Balancing mandatory overall operational ratings with optional clinical sub-ratings.

17. Security and Data Integrity

- **Password Hashing:** All user passwords are encrypted using bcrypt prior to database storage.
- **Stateless Authorization:** Secure JWT access tokens validated on all protected endpoints.
- **Request Validation:** Strict runtime validation using Zod schemas before data touches controllers or database.
- **Auditability:** Sensitive operations and security events logged to SystemLog with masked IP addresses.
- **Database Guardrails:** Foreign keys, unique indexes, and enums enforce strict structural integrity.

18. Future Enhancements

- **Automated Proof Verification:** Integration of OCR to automatically read and verify uploaded medical bills or receipts.
- **Interactive Frontend:** Dedicated React/Next.js web and mobile applications interfacing with the API.
- **Advanced Analytics Dashboard:** Visual graphs and reporting suites for healthcare admins.
- **Notification Engine:** Instant email/SMS alerts notifying users when their review status is updated.
- **Specialty & Location Search:** Full-text search and geofenced filtering by proximity and specialty.

19. Conclusion

MedReview addresses healthcare review manipulation through a structured relational database backend that couples proof verification with multi-dimensional rating metrics. Built with Node.js, Express.js, Prisma, and PostgreSQL, the platform establishes referential integrity across 12 models while offering role-restricted administration, community engagement tools, and security auditing. The design provides a trustworthy platform for healthcare decision-making.